Economics Letters 241 (2024) 111810

Contents lists available at ScienceDirect

Economics Letters

journal homepage: www.elsevier.com/locate/ecolet

Cognitive and mnemonic abilities in a trust game

Luca Delle Foglie [a], Stefano Papa [b,*]

[a] Department of Economics and Law, Sapienza University of Rome, Italy
[b] Department of Economics and Finance, University of Rome, Tor Vergata, CIMEO, Italy

ARTICLE INFO

JEL codes:
A13
C91
D03
D64
D90

Keywords:
Ambiguity attitude
Cumulative information flow
Mnemonic abilities
Cognitive Bias

ABSTRACT

Individual mnemonic abilities can play an important role in determining individual choices, especially in games that involve numerous decision-making situations in which the consistency of decisions depends on the information set retained by the subjects. In similar contexts it would be useful to distinguish the impact of cognitive abilities from mnemonic abilities on choices. We provide empirical evidence on the importance of providing cumulative information flow in games of the type described above.

1. Introduction

By using a measurement method that separates ambiguity attitudes from first-order beliefs in a trust game, Li et al. (2019) show that subjects' ambiguity attitudes and beliefs both matter for their trust decisions. Their method consists of 24 decision situations designed to elicit their matching probabilities for single and composite event. However, in the decision phase between ambiguous or risky prospect options, they are not provided any cumulative flow of information regarding the choices previously taken. Without cumulative information flow, the ability of experimental subjects to pass monotonicity tests may depend not only on cognitive abilities, but also on their mnemonic abilities. Following Battigalli and Generoso (2024), we add a cumulative information flow regarding the previous choices made to separate the cognitive abilities from the mnemonic ones, in the same strategic context of Li et al. (2019).

Section 2 presents the game we explore and the hypothesis we test. Section 3 describes our design. Section 4 reports results. Section 5 offers concluding remarks.

2. Game and hypothesis

In the trust game of Li et al. (2019), the trustor faces a binary choice. If she chooses the distrust option, both she and her trustee receive €10 for sure and there is no uncertainty. If trustor chooses the trust option, the outcome is uncertain because the amount trustor receives depends on the trustee's choice. The latter can choose among three feasible options: 1) R where both earn 15 euro each; 2) M where trustor earn 10 euro and trustee 18 euro,[1] and 3) S where trustor earn 8 euro and trustee 22 euro. (Fig. 1).

The three options R, M and S are the exhaustive and mutually exclusive ones that they call single events. While a composite event is the union of two single events such as: R or S; R or M; M or S. For each single or composite event, to elicit first-order beliefs and separate them from their ambiguity attitudes, participants in the role of trustor have to choose among:

a) Option 1 that pays 15 euro if the trustee chooses, for example, the single event R, and 0 otherwise **(ambiguous prospect);**
b) Option 2 that pays 15 euro with probability q and 0 with probability $1 - q$ **(risky prospect).**

We thank Pierpaolo Battigalli and Giancarlo Spagnolo for their valuable comments.
* Corresponding author.
E-mail addresses: luca.dellefoglie@uniroma1.it (L. Delle Foglie), stefano.papa@uniroma2.it (S. Papa).
[1] Option M is a selfish choice of trustee without hurting the trustor, but at a slight efficiency cost.

https://doi.org/10.1016/j.econlet.2024.111810
Received 4 April 2024; Received in revised form 23 May 2024; Accepted 10 June 2024
Available online 14 June 2024
0165-1765/© 2024 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

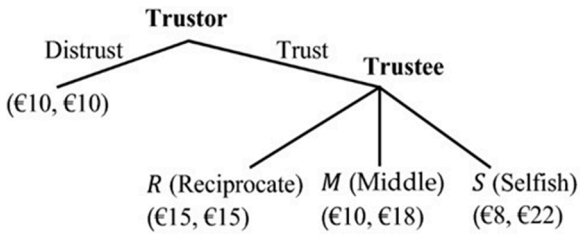

Fig. 1. Li et al. (2019)'s game.

Li et al. (2019) defines the matching probability of a single or composite event the probability q* such that a trustor is indifferent between the above ambiguous prospect and risky prospect with winning probability equal to q*. Therefore, it depends on her subjective first-order belief on the probability of realization of the event, for example R, but also on her ambiguity attitude.

If trustor chooses the ambiguous (risky) prospect, i.e. Option 1(2), the winning probability in the next following decision is increased (decreased). For each event, participants faced four decision situations, where Option 1 remained the same and the winning probability in Option 2 varied depending on the previous choices taken by participant.

Li et al. (2019) Fig. 4 shows how the events' matching probabilities are determined by subjects' choices in each block. Subjects faced 6 randomly ordered blocks (one for each single and composite event) of 4 decision situations each, therefore they faced a total of 24 decision situations regarding ambiguous and risky prospects. Based on the choices, the matching probabilities of each event and the first order beliefs corrected for the ambiguity attitude are estimated.

By monotonicity test definition of Li et al. (2019): *"Subjects' matching probabilities for a composite event should not be lower than those of the single events included in the composition. It is worth noting that for each subject were performed six monotonicity tests"*. In the event that a subject overestimates the probability of a single event more than once compared to the probability of a respective composite event, for example mp(R) > mp (R or M), it is not possible to estimate the aforementioned first-order beliefs and this involves the exclusion of the subject from subsequent analyses.

Battigalli and Generoso (2024) claim that memory is a subjective attribute of players and that their choices during a game are also affected by how they encode and store the information flow received during the game. Historically it has been assumed that subjects were characterized by perfect recall (e.g. Kuhn, 1953; Selten, 1975), but they suggest that the set of information flows that are consistent with what is retained may instead be determined by different mnemonic processes, such as Bounded, or Statistical, or Range memory. Considering the game of Li et al. (2019) in light of the above, it seems quite likely that the mnemonic abilities of the subjects, in a context characterized by many decision-making situations without any feedback and reminders on what was done previously, can play an important role in the subjects' ability to pass monotonicity tests. In an experiment like this it is not a trivial detail, since based on these tests the inclusion (exclusion) of the subject from the data analysis is determined, which should occur only for the criterion chosen by the experimenter and not also for other subjective characteristics.

Therefore, we differ from Li et al. (2019) by adding the cumulative information flow to reduce mnemonic bias to zero. Following Battigalli and Generoso (2024), we hypothesize that if subjects were provided with a reminder reporting the cumulative information flow, the percentage of them failing monotonicity tests would be lower than if it were not provided to them.

H1: proportion of subjects failing 2 or more monotonicity tests in the No Info treatment is greater than the proportion of subjects failing 2 or more monotonicity tests in the Info treatment.

Support for H1 would be consistent with our idea that mnemonic abilities may be a factor that can lead experimental subjects to make rationally inconsistent choices.

3. Experimental design[2]

The experiment involved 170 undergraduate students from the University of Rome Tor Vergata who participated in a single experimental session. On aggregate, we conducted 8 sessions, 4 sessions for the No Info treatment and 4 sessions for the Info treatment for a total of 78 and 92 participants, respectively. The experiment was programmed with the z-Tree software (Fischbacher, 2007).

More precisely, each session implemented the following sequence of stages, which was described to all players in the instructions before the experiment started:

1. **Trustor decision** Participants chose which of the following two options to implement if this choice was selected to determine their own and their partner's payments:
   a. Option 1: implements the partner's choice (i.e., the partner's Trustee decision);
   b. Option 2: pay 10 euros each.
2. **24 decision situations to elicit matching probabilities** Participants made their choice between the ambiguous prospect and the risky prospect for each single and composite event in blocks of 4 decisions each. In each block we referred to the same single or composite event and the probability of winning in option 2 varied depending on the choice made by the participant in the previous choices belonging to the same block (as described in Section 2).
3. **Trustee decision** Participants chose which of the following three options to implement if this choice was selected to determine their own and their partner's payments:
   a. Option A: pay 15 euros each;
   b. Option B: pay 10 euros to the Trustor and 18 euros to the Trustee;
   c. Option C: pay 8 euros to the Trustor and 22 euros to the Trustee.

The experiment as described so far, called No Info treatment, replicates the experimental design of Li et al. (2019), except for moving the survey to before the start of the experiment.[3] The Info treatment is distinguished from the No Info treatment by a single variation, namely that during the 24 decision-making situations between the ambiguous and the risky prospect the subjects were provided with a reminder on the screen containing information regarding the last choice made in each block of 4 choices previously addressed. Each subject was reminded of the actual options available for that specific decision-making situation, therefore the event to which Option 1 referred and the probability of winning in Option 2, and which of the two options she chose. This reminder was updated during the experiment and contained 5 reminders during the last block of 4 decision-making situations.

As regards payments, for each participant we randomly selected with equal probability one of the 26 choices made by the subject, i.e. the 24 choices between ambiguous and risky prospects, the trustor decision, and the trustee decision.[4] Each participant also received a fixed show-up fee of 2 euro.

[2] For the sake of brevity, the experimental design is just sketched. Instructions are available upon request

[3] They asked a survey after each block and before the subsequent one.

[4] If the trustor (trustee) decision was selected for a participant, the trustee (trustor) decision was selected for the respective partner, to guarantee the effective implementation of all the consequences deriving from the choice made by each partner.

4. Experimental results

In the Table 1 we report that in the No Info treatment 19 out of 78 participants failed 2 or more monotonicity tests (24.36% of participants), while in the Info treatment 11 out of 92 subjects failed 2 or more monotonicity tests (11.96% of participants). By carrying out a two-tailed 2-sample test for equality of proportions we find that the proportion of subjects who fail 2 or more monotonicity tests in the two treatments are different ($Z = 2.11$, $p = 0.035$), which constitute evidence in support of H1.

Table 1
Info and no Info: probability to pass the monotonicity test.

	Probability to pass	
	(a) Success	(b) No success
(1) Cumulative information flow	88.04 %	11.96 %
(2) No-Cumulative information flow	75.64 %	24.36 %

5. Concluding remarks

In an experimental game in which participants must face many decision-making situations, we find evidence that students can make inconsistent decisions (which in an experiment such as the one conducted would result in the exclusion of subjects from the analyzable sample) not only because of their cognitive abilities, but also because of their mnemonic abilities. We also find that, in the analyzed context, providing a reminder of previous choices significantly reduces the percentage of subjects (from 24% to 12%) who make inconsistent decisions.

By adding a cumulative information flow, during each decision-making situation, we have shown that we can avoid the exclusion of 12% subjects from the analyzable sample which has an impact on reducing the cost of research equal to approximately 327 euros. Finally, we believe it could be expected that the extent of the influence of mnemonic abilities on subjects' choices depends on the complexity of the experiment. Therefore, in experiments characterized by a greater number of decision-making situations, this effect could be even greater than that found by us.

Data availability

Data will be made available on request.

References

Battigalli, P., Generoso, N., 2024. Information flows and memory in games. Games. Econ. Behav. 145, 356–376.
Fischbacher, U., 2007. z-Tree: Zurich toolbox for ready-made economic experiments. Exp. Econ. 10, 171–178.
Kuhn, H.W., 1953. Extensive games and the problem of information. Contrib. Theory Games 2, 193–216.
Li, C., Turmunkh, U., Wakker, P.P., 2019. Trust as a decision under ambiguity. Exp. Econ. 22, 51–75.
Selten, R., 1975. Reexamination of the perfectness concept for equilibrium points in extensive games. Int. J. Game Theory 4, 25–55.